MATH317 · NUMERICAL METHODS

Homework 1 تدريب

Taylor · Horner · الأخطاء

محارب

01 الأسئلة

حل كل سؤال في المساحة اللي تحته. الحل الكامل في ملف منفصل.

Q1 RELATIVE ERROR

A real number x is represented approximately by 0.6032, and the relative error is at most 0.1%. What is x?

ROUNDING **Q2**

What is the relative error involved in rounding 4.9997 to 5.000?

MACLAURIN **Q3**

Derive the Maclaurin series of $(1 + x)^n$. Write it for n = 2, n = 3 and n = ½. Then use it to compute √1.0001 correct to 15 decimal places.

NUMBER OF TERMS **Q4**

How many terms of the series for $\ln(1 + x)$ are needed to compute $\ln 1.1$ with an error less than $\frac{1}{2} \times 10^{-8}$?

TAYLOR **Q5**

Write the Taylor expansion of $f(x) = x^3 - 2x^2 + 4x - 1$ in the form $f(2 + h)$.

COMPLETE HORNER **Q6**

Use the complete Horner algorithm to find the Taylor expansion of $p(x) = 3x^5 - 2x^4 + 15x^3 + 13x^2 - 12x - 5$ about $r = 2$.

COMPLETE HORNER **Q7**

Find the Taylor series of $f(x) = x^4 - 4x^3 + 7x^2 - 5x - 2$ about $c = 3$.